scientific reports

OPEN

Disturbance event impact on aboveground carbon storage and vulnerability of large trees in old-growth coniferous forest stands in Latvia

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Large, old trees store a considerable part of the total carbon in old-growth forests; however, their influence on long-term climate change mitigation capacity remains unclear. Damage caused by wind and insect outbreaks can rapidly kill trees, and these natural events are expected to increase with climate change, thereby reducing the long-term carbon storage capacity of old-growth stands. The aim was to estimate the importance of large, old trees in carbon storage at the stand level and the potential risks for the carbon stock if they die. The data were collected from Scots pine and Norway spruce stands on mineral soils, of which 44 were old-growth stands, and 47 were mature stands two times younger than the former. Stand-level carbon storage in tree biomass was investigated; the presence of deadwood was analyzed. The carbon stock changes were calculated by excluding the 1–15 largest trees per sample plot. The results showed that the largest trees in old-growth coniferous stands hold a disproportionally large (~ 50% of the total tree biomass carbon pool) share of carbon compared to mature stands, ~ 50% and ~ 10%, respectively. A reduction of the tree biomass carbon pool by approximately 50% required a removal of around 30% more trees in mature than in old-growth stands. A relatively low proportion of deadwood from the total stand volume (~ 20%) suggests a limited single-tree mortality and vulnerable mitigation potential in the assessed old-growth stands. Nevertheless, continuous single-tree mortality due to ageing or windthrow re-shapes forest structure, and raises the potential risk of the stands to maintain and/or further increase substantial tree biomass-linked carbon pools.

Keywords Stand density, Carbon stock, Deadwood, Stand fragility, Hemiboreal region

Forests are the key terrestrial ecosystems in northern Europe, but they have largely been strongly transformed by past anthropogenic activities through intensive exploitation and modern forestry, which have altered their natural dynamics and structure[1,2]. As a result, old forests with natural structures are rare and unevenly distributed, mostly occurring in regions that were difficult to access or persisting as remnants of virgin forests, but these few remaining areas provide a unique opportunity to study the dynamics and carbon-carrying capacity of old-growth forests for the future[3,4]. Additionally, remaining primary and old-growth forests in Europe only partially represent the full range of environmental conditions within the region, and many data gaps remain[5]. It has been reported that approximately 4.9 million hectares of the European Union (EU) forests are classified as primary or old-growth forests, covering an estimated 3% of the European forest area[3]. However, due to the lack of a homogeneous definition, the estimated area in the EU fluctuates across different sources, ranging from 1.5 million hectares[6] to 3.7 million hectares[5]. The term 'old-growth' refers to a wide diversity of forest ecosystems in terms of stand age and structure, tree species composition, broad variation in tree size, as well as the spacing of live trees and deadwood of various sizes and decay stages, both across regions and sometimes even within a region[7–10]. Moreover, due to the long history of intense forest management and land use in Europe, the term "old-growth forests" is widely applied to areas unlogged for at least 40–80 years that have regained many old-growth features, naturalness, and ecological function, without implying the absence of past human disturbance[11].

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The carbon dynamics in a forest ecosystem comprise the carbon uptake by trees, and after forest productivity reaches its maximum, it declines with stand ageing, resulting in a decrease in the carbon storage[2,4,12]. Tree biomass has been found to be the greatest carbon pool in unmanaged stands, but after reaching the plateau stage, whose length depends on tree species and disturbance regime[13–15]. Numerous studies have shown that old-growth forests can serve as long-term carbon reservoirs, particularly under stable forest conditions characterized by low disturbance frequency and limited disturbance synchrony, although their carbon accumulation rates are generally lower than those of younger, managed stands[11,16]. The ecosystem's carbon-carrying capacity represents the potential stock that can be sustained by natural processes, including disturbances; however, intensified disturbance regimes are likely to impact carbon storage[17]. When disturbances occur, forest stands can temporarily shift from being net carbon sinks to net carbon sources. Therefore, the sustainability of forest carbon sinks largely depends on how rapidly forests recover their photosynthetic capacity following such events, as well as management strategies of these forests aimed at maintaining vertical complexity and special heterogeneity[11,18,19]. Over time, the old-growth forests' ability to store large amounts of carbon in tree biomass and soil remains in a fragile balance between growth and mortality[2,20,21].

Wind and biotic attacks (insect outbreaks, fungal pathogens) are a mode of natural disturbance – distinct from human-made disturbances – that represents a primary driver of forest structure and a key factor influencing deadwood carbon pool dynamics, potentially leading to a substantial reduction in forest carbon stocks[11,22,23]. In the northern European region, non-stand-replacing disturbances such as small-scale tree mortality, including gap dynamics, are the dominant driver of tree mortality in forest stands[1]. Tree mortality is one of the key processes shaping the dynamics of unmanaged and old-growth forests, influencing successional pathways and community composition, and creating gaps that facilitate regeneration[1,24]. Numerous interacting processes contribute to tree mortality, and to better categorize its diverse causes, the concept of modes of mortality has been applied, encompassing both tree-related factors (competition, vitality, age, species, etc.) and natural disturbance processes[24–26].

It has been well recognised that an important component of old-growth forests is the large amount of deadwood they possess within the entire spectrum of decomposition stages, also storing carbon, potentially mitigating climate change, and maintaining biodiversity[26]. Deadwood is a key component of nutrient cycling, and it plays an important role in maintaining forest structure and functions, and it represents a unique biodiversity reservoir, therefore, playing an important role in biodiversity conservation[23]. The latest research shows that disturbances in forests, which have been strongly increasing over the past decades, pose the highest risk of losing forest multifunctionally in western Europe[27,28]. These findings emphasise the need for integrating specific carbon management objectives within a comprehensive forest management framework, both to sustain ecological functions and climate mitigation potential[18].

Previous studies in the European Mediterranean and also hemiboreal forests have found that the maximum accumulation of live biomass and, therefore, carbon storage efficiency can be reached in mature stands and it lowers efficiency over the time compared to old stands[21,29,30]. In addition, differences that have been revealed between stand density in old-growth and younger stands, suggested that the few largest trees store a considerable part of the total tree carbon[29,30]. This means that the death of large trees in old-growth stands (i.e. the stands where the old trees are the dominant forest element), which is considered one of the most serious consequences of natural tree dieback from ageing or disturbances[26,31], may cause a dramatic decline in the amount of carbon stored in tree biomass[16,28]. Single-tree mortality caused by disturbances is a complex process that is influenced by stand characteristics (structure, age, species composition), and the history of management[32], as well as the presence of secondary biotic agents[33]. A study in Latvia shows that the survival rate of Norway spruce stands decreased with increasing stand age, especially in stands after the 6th decade[32]. In areas with a low influence of disturbances, primary and old-growth forest stands ensure substantial carbon storage; here, the presence of large trees is the most important factor for determining high carbon storage[17].

Here, we took advantage of the previously measured old-growth stands and two times younger mature stands in hemiboreal Latvia to estimate how vulnerable such stands actually are, and how this could reduce the old-growth forest stands' true mitigation potential, as carbon storage dynamics depend heavily on tree size and stand density according to previous studies. The main aim was to analyze how single-tree mortality could impact tree carbon storage by comparing mature and old-growth Scots pine (*Pinus sylvestris* L.) and Norway spruce (*Picea abies* L. Karst.) stands on minerals soil with a normal moisture regime. In particular, we tested the hypothesis that the largest trees in old-growth coniferous stands hold a disproportionally large share of carbon compared to mature stands and that there is a potential risk for the carbon stock reduction due to their death because of the dieback, windthrown or other factors Therefore, an assessment of the potential impact of large tree carbon storage can provide a valuable insight that can be used for further discussions and in the development of recommendations for different management strategies in old-growth stands.

Materials and methods

Study area and stand selection

Latvia is located in the European hemiboreal forest zone[34], which is a transitional zone including south central Sweden, the southern part of Finland, the Baltic states, and parts of neighboring Poland, Russia and Belarus, hosting a variable mixture of coniferous and deciduous trees and a diverse soil and biota[35,36]. Overall, coniferous trees are found on infertile soils and deciduous trees on more fertile soils[36]. The Latvian National Forest Inventory (NFI) data show that 55% of Latvia is covered by forest (3.6 M ha forest land), and the two most common forest types are *Hylocomniosa* and *Oxalidosa*, covering 40% of the total forest area. According to the national classification[37], both forest types are associated with mesotrophic mineral soils with a normal moisture regime: *Hylocomniosa* has dry, moderately nutrient-rich mineral soil, and *Oxalidosa* has dry, nutrient-rich mineral soil. Both forest types include stands dominated by Scots pine, Norway spruce, and both species coexist naturally due

to overlapping natural distribution and growth patterns[38]. The climate is cool and moist, influenced by the Baltic Sea and the North Atlantic. According to data from the Latvian Environment, Geology and Meteorology Centre, the mean annual sum of precipitation is 685 mm, and the mean annual air temperature + 6.8 ℃.

Old-growth forests were defined based on the Buchwald[7] classification, but in this study specific subgroups of old-growth forest stands, in which old target-species trees still formed the dominant cohort were analyzed. A number of criteria (such as size, location, former management activities, stand age) were used to preselect potential old-growth stands from the forest inventory database. Subsequently, stands were visited on-site and their compliance with the criteria was assessed. In 2016 and 2017, measurements were done in 19 Norway spruce stands (170 to 205 years old) and 25 Scots pine stands (163 to 218 years old)[39,40] (Fig. 1). In total, 44 old-growth stands with 275 sampling plots (6 to 8 sample plots in each stand) of 500 m^2 (radius 12.62 m) were regularly established (Table 1).

To establish reference and to adapt forest management strategies, we compared old-growth forest stands with managed mature stands to assess how tree mortality patterns differ between these forest types. In this study mature stands from multifunctional forests, characterizing younger mature stands were sought as close as possible to the measured old-growth stands (Fig. 1).

Potential mature stands were preselected from the forest inventory database, based on the following criteria: (1) a distance no more than 5 km from the corresponding old-growth stand; (2) stands dominated by the target species; (3) species composition corresponding to nearest old-growth stand; (4) an age of 75 to 100 years for pines, 70 to 100 years for spruces; (5) *Hylocomniosa* or *Oxalidosa* forest type. After the search process, mature stands were measured using the same methodology in both 2021 and 2022 (Table 1). In total, 37 mature spruce stands and 10 mature pine stands were selected, altogether 188 sample plots were regularly established (4 sample plots, 500 m^2 each, radius 12.62 m) (Fig. 1; Table 1).

Data collection

A full description of the fieldwork methodology, which was applied in both old-growth and mature stands, is published in Ķēniņa et al.[29]. In circular sample plots (500 m^2; radius 12.62 m), the tree species and tree positions (relative coordinates) were recorded for live and dead trees. The diameter at breast height (DBH) of all living trees (DBH ≥ 6.1 cm) and tree height (DBH ≥ 6.1 cm) in the first and second layers for selected trees were measured. The tree DBH and tree height of each standing dead tree (with or without the top (snags), DBH ≥

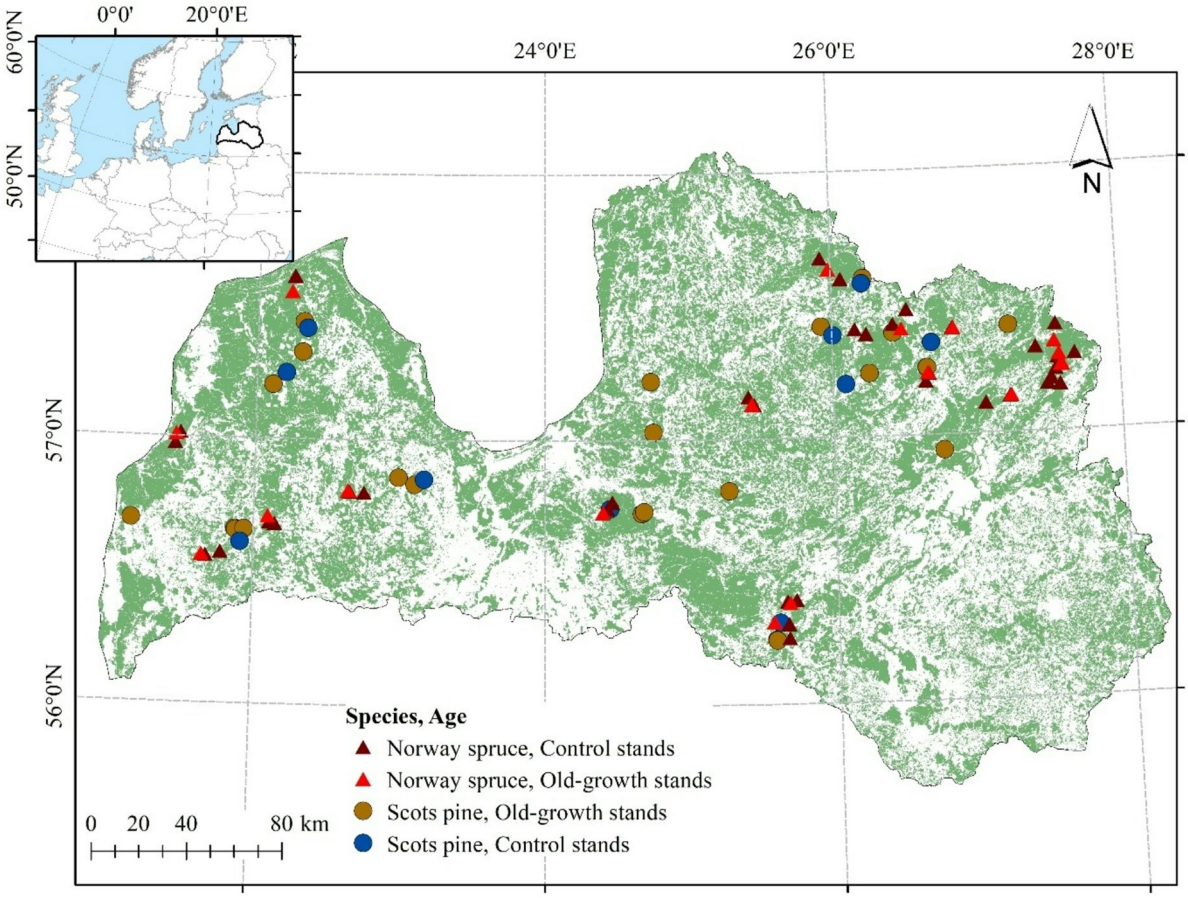

Fig. 1. Location of measured old-growth and mature Scots pine and Norway spruce stands on mineral soils in hemiboreal Latvia. Maps were produced using ArcMap 10.5 (ESRI, Redlands, CA, USA).

Characteristics	Spruce		Pine	
	Old-growth	Mature	Old-growth	Mature
Mean stand age, years	182 ± 11.0	77 ± 6.2	179 ± 15.6	84 ± 6.6
Studied stands in *Hylocomniosa*	8	17	25	10
Studied stands in *Oxalidosa*	11	20	-	-
Total number of studied stands	**19**	**37**	**25**	**10**
Total number of sample plots	111	148	164	40
Total number of sampled trees	5176	5831	6783	1397
Total number of sampled large (DBH > 40 cm) trees	426	200	1044	48
DBH range of sampled trees	2.1–77.0	2.1–72.1	2.1–74.8	2.1–52.5

Table 1. Characteristics of sampled stands. Sampled stands categorized by dominant tree species and stand type – old-growth and mature stands. Mean (± standard deviation) stand age.

6.1 cm) were also measured. For lying deadwood, the length and diameter at both ends (≥ 14.1 cm at the ticker end) was measured. In the center of the sample plot, a smaller subplot (a quarter circle of 25 m^2) was established, where living and dead trees of smaller diameters were recorded (DBH 2.1–6.0 cm for standing trees and 6.1–14.0 cm diameter for lying deadwood). For all dead trees, we noted tree species and decay stage according to the five-class classification by Sandström et al.[41]. Stand age was obtained from forest inventory data, and in each stand, the age of dominant trees of the dominant tree species was assessed using 12 increment cores.

Tree height for all living trees was expressed as a function of tree DBH using Näslund's model. The individual tree volume was calculated based on local equations using tree DBH and tree species[42]. The individual tree biomass was calculated using tree DBG and height based on local allometric equations for above- and belowground biomass components of major tree species in Latvia[43,44]. Above-ground biomass (kg) was calculated by summing the stem biomass and branch biomass, whereas the belowground biomass was calculated by summing the coarse root, stump and small root biomass, as in the following equation:

$$Tree\,biomass = k \times exp\left(a + b \times \left(\frac{DBH}{DBH + m}\right) + c \times H + d \times ln\,(H)\right), \quad (1)$$

where DBH is diameter at breast height (cm); H is tree height (m); a, b, c, and d are regression parameters; m is a constant; k represents a correction factor[43,44].

For *Betula* sp. belowground biomass and Scots pine small root biomass, as well as for grey alder (*Alnus incana* (L.) Moench) and common alder (*Alnus glutinosa* L.) belowground biomass and small root biomass, modified equations were used[43,44]. The Eq. 1 of the tree species with the most similar wood density was used to calculate biomass for the species that do not have such models (e.g., *Quercus robur* L., *Tilia cordata* Mill., *Sorbus aucuparia* L., *Corylus avellana* L.). For converting tree biomass into the living tree biomass carbon stock, species-specific carbon content values developed in Latvia and recommended for reporting for the main tree species in Latvia were applied[45].

The volume of dead standing trees (m^3) was calculated based on DBH and tree species, in accordance with Liepa[42]:

$$V = \psi \times L^{\alpha}\, DBH^{\beta\, lgL + \phi}, \quad (2)$$

where L is stem length (m); DBH is tree tree diameter at breast height (cm); ψ, α, β, $and\ \phi$ is coefficients for tree species[42].

The volume of the snags and the lying dead wood (m^3) was calculated using Huber's formula[25]:

$$V = \frac{L\pi\, {d_m}^2}{4}, \quad (3)$$

where L is snag height/log length (cm); d_m is mid-diameter of log or snag (cm).

For deadwood carbon stock calculations, volume (Eq. 2 or Eq. 3) was converted to mass using decay stage-specific wood density, and carbon content values were used as applied in the study by Köster et al.[46] in Estonia.

Data analysis

The data analysis was performed using R version 4.4.0[47]. The R libraries lme4[48] and lmerTest[49] were used to implement the linear mixed-effects models (LMER).

The R library emmeans[50] was used to calculate and compare the estimated marginal means. The R library forestSAS[51] was used to calculate the uniform angle index (UAI). All tests were performed at α = 0.05. The forest type (*Hylocomniosa* or *Oxalidosa*) impact was assessed using an LMER test – because carbon stock does not significantly differ between forest types, this factor was not evaluated further. A Chi-Square test was performed to evaluate the differences between DBH classes and the number of trees among stand types.

LMER models (Eq. 4) were used to test the effect of stand type (spruce or pine of mature and old-growth stands) as an independent variable on the dependent variables – the stand characteristic variables, and the carbon stocks in the live tree biomass:

$$y_{ij} = \mu + (T_{ij}) + \varepsilon_{ij}, \quad (4)$$

where y_{ij} is the value of analysed parameter of particular stand; μ is overall mean value for analysed parameter; T_{ij} is the effect of forest type; (Y_{ij}) is the random effect (intercept) of a stand, ε_{ij} is the error term.

Each stand's characteristic variable had its own model. The stand identification was used as a random factor, as there were multiple plots per stand. A Tukey's test was used as a Post Hoc test to compare the estimated means between stand types. To investigate whether and how largest tree exclusion affects the tree biomass carbon stock, at a sample plot level, the total tree carbon stock for each sample plot we calculated and then recalculated by excluding between 1 and 15 largest trees per sample plot. Then we fitted separate LMER models for spruce and pine to assess carbon changes and their percentage change based on the number of excluded trees as a continuous variable (1 to 15 trees per sample plot). Afterwards, a comparison between age groups (mature, old-growth) according to the amount of the excluded trees was performed. The stand identification was used as a random factor.

The spatial structure of the sampled deadwood was characterized by the spatial distribution of dead trees' positions (including all deadwood types – for lying deadwood the largest end's position was included) using the uniform angle index (UAI)[52]. UAI for deadwood was calculated for each sampling plot with more than 10 deadwood units. Also, spatial distribution of live trees (first and second layer) within the sample plot was characterized by using UAI. To eliminate edge effects, UAI for trees in the buffer area was not calculated. A one sample T-test was used to compare the mean uniform angle index values for each sample plot with 0.5 to determine if random or clustered distribution is present. The standard angle (72°) was applied.

Results

Stand-level carbon storage in tree biomass and deadwood

The stand characteristics of the old-growth stands were significantly different from those found in mature stands (Table 2). The old-growth stands (although pine was significantly larger than spruce, 40.9 ± 4.1 cm and 35.8 ± 4.8 cm, respectively) had a higher 1 st layer mean DBH compared to mature stands, where it was similar between species. We estimated significantly higher mean stand parameters in old-growth pine stands, such as stand volume (568 ± 104 m^3 ha^{-1}) and basal area (45.2 ± 6.2 m^2 ha^{-1}). There were non-significant differences in stand deadwood volume between the old-growth pine and spruce stands. Additionally, much lower but similar stand density in the first layer was observed in old-growth pine (272 ± 102 trees ha^{-1}) and spruce (274 ± 119 trees ha^{-1}) stands compared to mature stands, where mature spruce stands (528 ± 87 trees ha^{-1}) had a significantly higher density compared to mature pine stands (408 ± 167 trees ha^{-1}) (Table 2).

The old-growth stands showed a much wider range of diameter classes at the stand level (Fig. 2). Large trees (DBH > 40 cm) were more frequently detected in old-growth stands compared to mature stands (Fig. 2), as also indicated by the significantly larger first-layer tree DBH of old-growth stands (Fig. 2). Overall, there was a significant difference (both $p < 0.001$) of tree distribution in diameter classes between old-growth and mature stands in pine and spruce stands.

In old-growth stands, large trees (DBH > 40 cm) comprised 22% and 14% of all inventoried trees (first and second layer) in pine and spruce stands, respectively (Fig. 2). In contrast, large trees comprised only 4% of all inventoried trees (first and second layer) in both pine and spruce mature stands. An analysis of the tree biomass carbon stock distribution through diameter classes revealed that large trees (DBH > 40 cm) in old-growth stands stored 58% and 49% of carbon in pine and spruce stands, respectively, while in mature stands, this was significantly lower – 11% in pine and 14% in spruce (Fig. 2).

	Spruce		Pine	
Stand parameters	Old-growth	Mature	Old-growth	Mature
Tree DBH, cm	18.3 ± 4.2 [a]	22.6 ± 3.0 [b]	22.6 ± 3.5 [b]	23.4 ± 5.9 [b]
Tree height, m	16.3 ± 3.3 [a]	22.3 ± 2.3 [c]	18.7 ± 2.8 [b]	22.1 ± 4.6 [c]
Stand volume, m^3 ha^{-1}	433 ± 121 [a]	444 ± 89 [a]	568 ± 104 [b]	425 ± 171 [a]
Stand basal area, m^2 ha^{-1}	36.4 ± 7.3 [a]	35.0 ± 5.4 [a]	45.2 ± 6.2 [b]	34.1 ± 10.4 [a]
Stand deadwood volume, m^3 ha^{-1}	89.3 ± 46.4 [b]	32.2 ± 35.2 [a]	95.5 ± 39.1 [b]	11.3 ± 67.7 [a]
Tree DBH 1 st layer, cm	35.8 ± 4.8 [b]	27.4 ± 3.5 [a]	40.9 ± 4.1 [c]	30.1 ± 6.7 [a]
Tree height 1 st layer, m	27.4 ± 2.9 [ab]	26.4 ± 2.1 [a]	29.3 ± 2.6 [b]	27.9 ± 4.1 [ab]
Basal area 1 st layer, m^2 ha^{-1}	28.4 ± 10.4 [a]	31.7 ± 7.8 [a]	36.5 ± 9.3 [b]	29.9 ± 5.9 [a]
Tree density 1 st layer, trees ha^{-1}	274 ± 119 [a]	528 ± 87 [c]	272 ± 102 [a]	408 ± 167 [b]
Stand volume 1 st layer, m^3 ha^{-1}	379 ± 119 [a]	418 ± 88 [a]	497 ± 103 [b]	393 ± 169 [ab]

Table 2. Estimated mean values (± confidence interval) of the stand parameters measured in the studied stands. Estimated mean (± confidence interval) tree diameter at breast height (DBH), height, basal area and stand density and stand volume of first-layer trees (1st layer). [a, b,c] different letters indicate groups with significant differences.

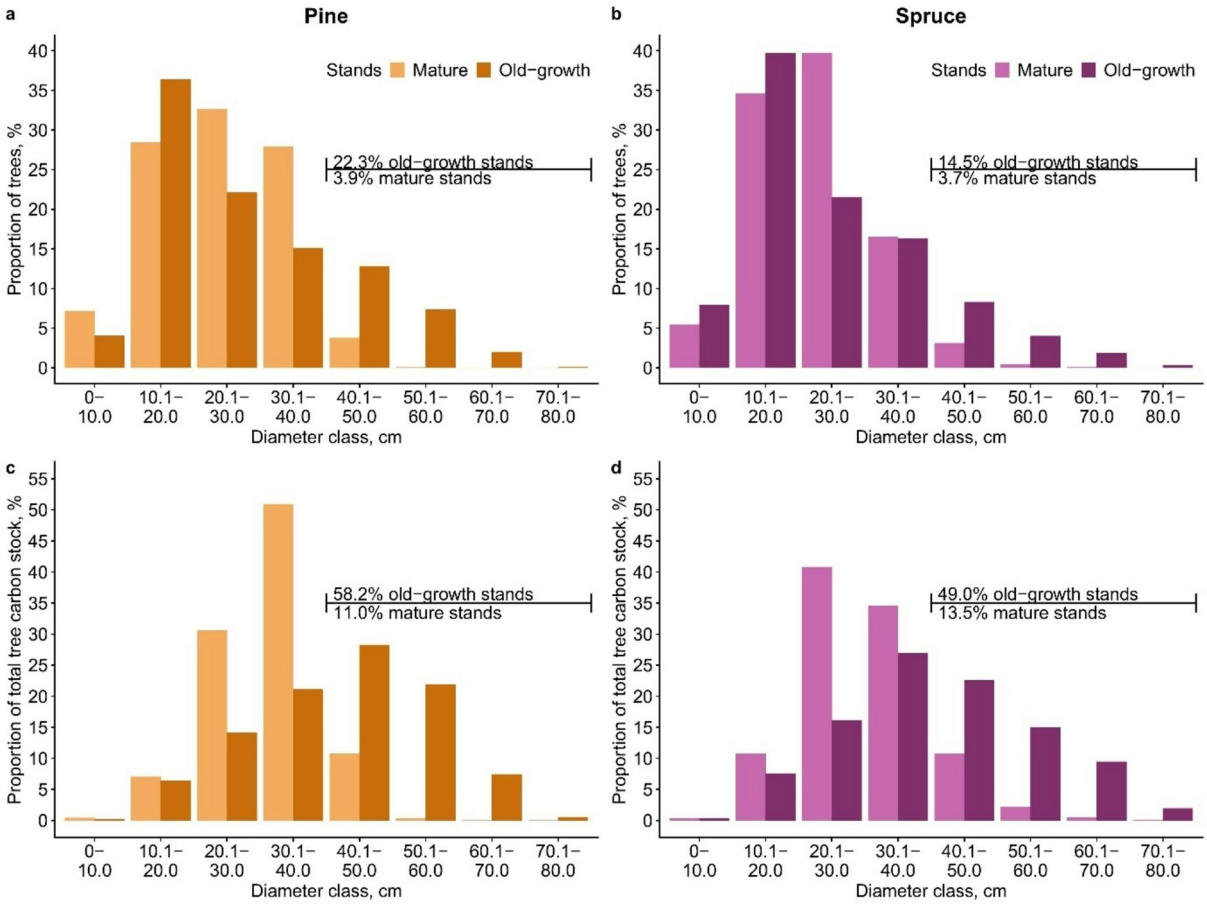

Fig. 2. Percentage of inventoried trees **(a, b)** and percentage of carbon stock (C t ha^{-1}) in tree biomass distribution **(c, d)** (first and second-layer trees) in diameter classes (from 10 to 80 cm) in spruce and pine old-growth and mature stands.

The estimated mean tree biomass carbon stock (± confidence interval) for old-growth pine stands (181 ± 32 C t ha^{-1}) was significantly higher than in old-growth spruce (148 ± 37 C t ha^{-1}) and mature stands (pine (146 ± 53 C t ha^{-1}) and spruce (154 ± 27 C t ha^{-1}) (Fig. 3). The carbon stock in tree biomass ranged from 63 to 300 C t ha^{-1} in spruce old-growth stands and from 44 to 313 C t ha^{-1} in mature spruce stands. In pine stands, the carbon stock in tree biomass ranged from 77 to 259 C t ha^{-1} in old-growth stands and 75 to 202 C t ha^{-1} in mature stands.

We did not find a significant difference in deadwood volume and deadwood carbon pool between old-growth pine and spruce stands (both $p > 0.05$). However, significant differences (all $p < 0.001$) appeared among the age groups. The estimated mean of the total deadwood carbon stock was 2.3 ± 2.52 C t ha^{-1} (± standard error) in mature pine and 5.12 ± 0.869 C t ha^{-1} in mature spruce stands, and 14.8 ± 1.30 C t ha^{-1} in old-growth pine and 12.11 ± 1.07 C t ha^{-1} in old-growth spruce stands. However, the ranges were highly variable in the old-growth stage for both pine and spruce, ranging from 0.5 to 74.6 C t ha^{-1} and 0.2 to 71.1 C t ha^{-1}, respectively. Notably smaller variation was observed in the mature stage, with pine stands ranging from 0.02 to 9.5 C t ha^{-1} and spruce stands from 0.02 to 28.5 C t ha^{-1}. For both spruce and pine, the carbon stock significantly differed between deadwood types in the mature and old-growth stages. In pine stands, the carbon stock in lying deadwood and standing dead trees was similar ($p > 0.05$) in both mature and old-growth stands. In spruce stands, the carbon stock in lying deadwood constituted the greatest proportion, 52% in mature and 61% in old-growth stands (Fig. 4a). However, in pine stands, although lying deadwood stores 38% of deadwood carbon in both mature and old-growth stands, most of the deadwood carbon was found in standing dead trees, 43% in mature and 49% in the old-growth stage (Fig. 4a). Snags carbon constituted the smallest proportion (from 11 to 19%) of the total deadwood carbon for all analyzed stands (Fig. 4a). In all stand types, most of the deadwood carbon was in decay stage 2 (weakly decayed), comprising 54% to 83% of the total deadwood carbon (Fig. 4b).

Tree-level carbon storage modelling in sample plots

At the sample plot level, we analyzed how the exclusion of the largest trees affects the tree biomass carbon stock and its percentage change based on the number of excluded trees (1 to 15 trees per sample plot) between old-growth and mature stands (Fig. 5). Average tree carbon stock rapidly increased with increasing tree DBH in

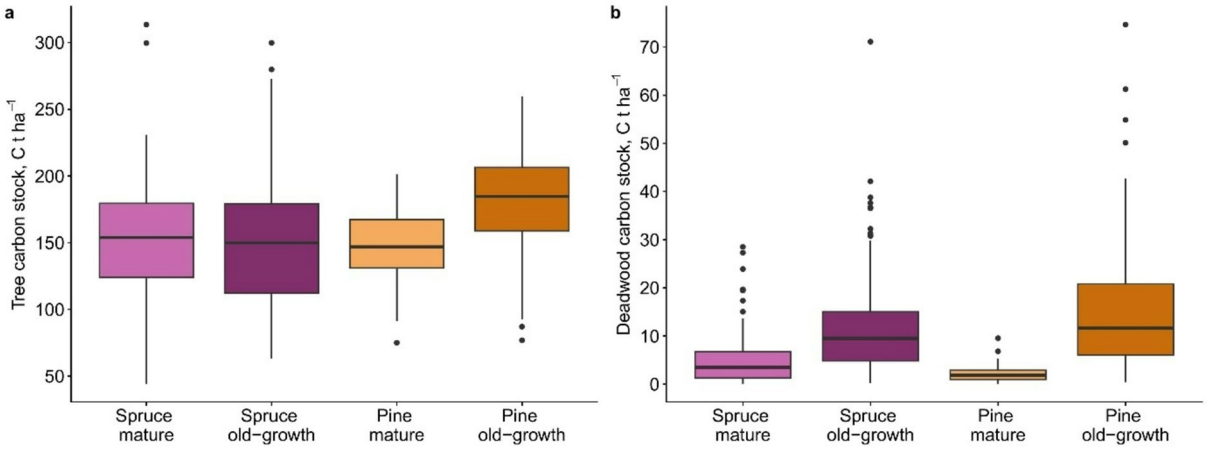

Fig. 3. Tree biomass **(a)** and deadwood **(b)** carbon stock box-plot of pine and spruce old-growth and mature stands. The box shows 25th and 75th percentiles, median (horizontal line) and whiskers show minimal and maximal values or values within 1.5 times the interquartile range if smaller/larger values present (points).

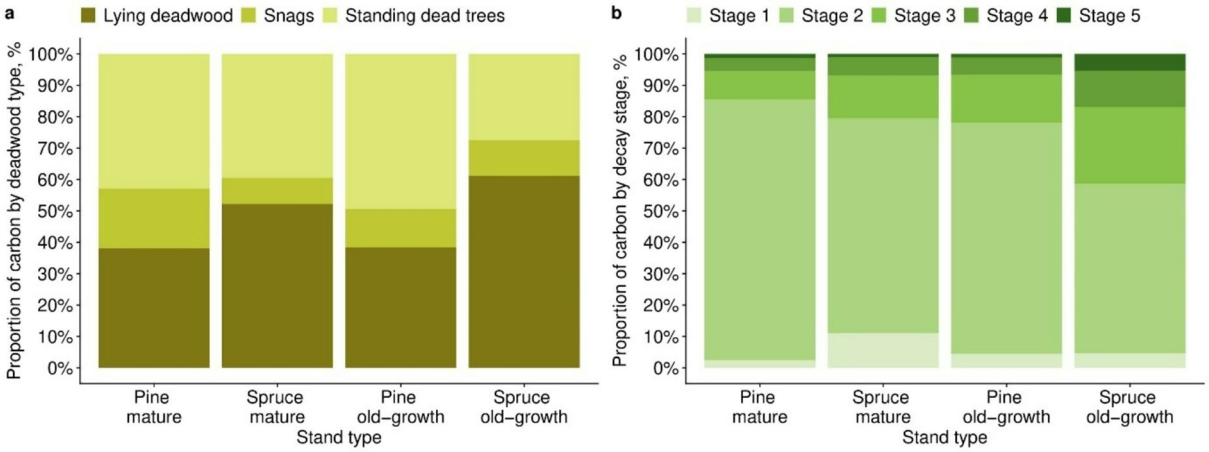

Fig. 4. Proportion of deadwood carbon stock in pine and spruce old-growth and mature stands according to **(a)** deadwood type and **(b)** deadwood decay stage. Decay stage was assessed using a five-class classification following Sandström et al.[41], where Stage 1 represents recently dead trees (sound wood) and Stage 5 corresponds to very decayed dead wood.

old-growth stands for all measured trees in assessed sample plots. However, the obtained results clearly showed that for large trees, a small increase in tree DBH corresponds to a big increase in carbon storage and there were important differences between species distribution between pine and spruce stands.

In pine stands, the mean initial tree carbon stock in the sample plot was 9.0 C t ha^{-1} and 7.3 C t ha^{-1} in old-growth and mature stands, respectively. The initial mean tree biomass carbon stock in sample plot was significantly ($p < 0.001$) higher in old-growth pine stands comparing to mature stands. However, the initial mean DBH of trees in the sample plot was similar between mature and old-growth stands (23.4 ± 2.7 cm and 22.6 ± 1.3 cm, respectively). After excluding eight trees, the mean DBH of the remaining trees decreased to 19.3 ± 2.4 cm in mature stands and 16.0 ± 1.0 cm in old-growth, showing a significant difference. This difference increased with the number of excluded trees, reaching 11.8 ± 0.8 cm and 15.6 ± 1.9 cm in old-growth and mature stands, respectively. (Fig. 5e). By excluding one to three of the largest trees from the sample plot, carbon storage in tree biomass was significantly different ($p < 0.001$) between old-growth and mature stage (Fig. 5a). After excluding more (four to fifteen) trees per sample plot carbon stock in the sample plot became similar between old-growth and mature pine stands. In old-growth pine stands on average one big tree stores around one ton of carbon. One to fifteen tree exclusion caused 12% to 87% and 9% to 79% tree biomass carbon stock reduction per sample plot in old-growth stands and mature stands, respectively. Percentage reduction of tree carbon stock per sample plot was higher in old-growth pine stands, and significant ($p < 0.05$) differences between mature and old-growth stands were observed when excluding two to fifteen trees per plot (Fig. 5c). Removing the six largest trees per sample plot caused tree biomass carbon reduction by 53% in old-growth stands and the exclusion of eight trees

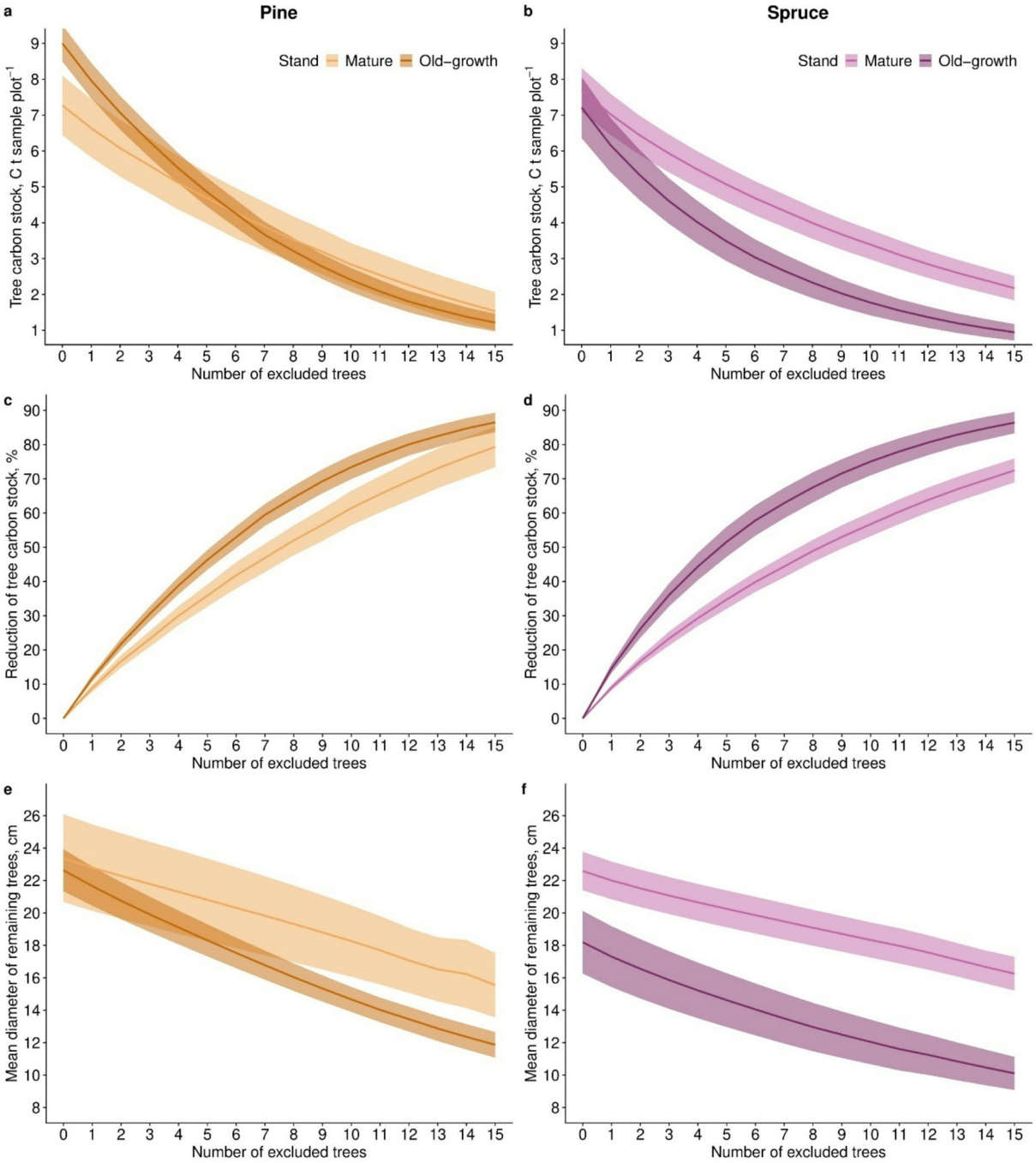

Fig. 5. Relationship between the number of excluded trees from the sample plot (500m^2) with **(a**, **b)** tree biomass carbon stock, **(c**, **d)** percentage of the tree carbon stock reduction, and **(e**, **f)** mean diameter at breast height (DBH) within the sample plot of pine and spruce old-growth and mature stands.

from the sample plot reduced the tree biomass carbon stock by 52% in mature stands of pine. Omitting the ten largest trees from the sample plot reduced the tree carbon stock by 61% in mature and 73% in old-growth pine stands.

In spruce stands, the mean initial tree biomass carbon stock in sample plots was similar in the old-growth (7.2 C t ha^{-1}) and mature (7.7 C t ha^{-1}) stages. Contrary to pine stands, when excluding one to fifteen trees per sample plot, the carbon stock in tree biomass between old-growth and mature stands was significantly ($p < 0.01$) different (Fig. 5b). The exclusion of one to fifteen trees caused 15% to 86% and 9% to 73% tree biomass carbon stock reduction per sample plot in old-growth stands and mature stands, respectively. Similar to pine stands, the percentage reduction of tree carbon stock per sample plot was higher in old-growth spruce stands, and significant ($p < 0.001$) differences between mature and old-growth stands were observed when excluding one to fifteen trees per plot (Fig. 5 d). The exclusion of five of biggest trees in old-growth spruce sample plots reduced

tree carbon stock by 51%. In mature spruce stands the exclusion of nine trees reduced the tree biomass carbon stock by 53%. The exclusion of the ten largest trees from the sample plot reduced the tree carbon stock by 57% in mature and 75% in old-growth spruce stands. In contrast to pine stands, the initial mean DBH of trees was significantly larger in mature stands (22.6 ± 1.2 cm) than in old-growth stands (18.2 ± 1.9 cm), and the difference increased with increased with the number of excluded trees, being more pronounced in old-growth stands, where the mean DBH decreasing to 10.1 ± 1.0 cm (Fig. 5f).

Spatial distribution of live and dead trees in sample plots of old-growth stands

The mean UAI of deadwood was 0.55 ± 0.01 in pine and 0.57 ± 0.01 in spruce old-growth stands, respectively. Deadwood UAI for both species significantly differ from 0.5: 78% and 87% of all deadwood units in pine and spruce old-growth stands exhibited a spatial aggregated pattern. The mean UAI of live trees was 0.52 ± 0.01 in pine and 0.53 ± 0.01 in spruce old-growth stands, respectively. The proportion of randomly distributed trees in sample plots was relatively low, with 43% and 38% of live trees in pine and spruce old-growth stands having UAI under 0.5, respectively, meaning that trees are more evenly spaced.

Discussion

Large trees as drivers of tree biomass carbon pool and estimation of deadwood carbon pool

Our hypothesis was confirmed – the largest trees in old-growth coniferous stands hold disproportionally large shares of carbon compared to two times younger mature stands. Boreal and also hemiboreal old-growth forests are specific and often undervalued ecosystems. They may present only a few of the structural attributes compared to such forests in the temperate forest zone[1]. Nevertheless, they play a critical role in mitigating climate change because of their high carbon storage capacity, they serve as biodiversity hotspots and also provide key ecosystem services such as water system and soil protection[1,8]. In our study, the carbon storage capacity of old-growth stands and two times younger mature coniferous stands was indicated by differences in stand parameters (stand and soil characteristics, carbon stock) (Table 2; Fig. 2).

The present study shows that stand density decreases with increasing tree size, and overall, tree density was much lower in old-growth stands, as has been reported in previous studies[30]. Additionally, a wider tree distribution in diameter classes was observed in old-growth stands, and comparing tree species – in pine dominated stands. Similar results have been obtained in uneven-aged Scots pine stands in central Poland[53], which may reflect differences in the species' biology (spruce is a shorter-lived species and is also more vulnerable to root rot and bark beetles outbreaks)[15,33,54]. The old-growth stands had higher 1 st layer mean DBH (although there were significant differences between pine and spruce stands) compared to mature stands, where DBH was similar between species. This was expected, because on one hand, younger stands grow faster, but on the other older bigger trees are still continuing to grow[55], therefore, large trees were more frequently detected in old-growth stands compared to mature stands. Previous studies and our study confirm that the few largest trees in old-growth stands store a considerable part of the total tree carbon[29,30]. Large trees (DBH > 40 cm) comprised 22% and 14% of inventoried trees in old-growth pine and spruce stands, respectively (Fig. 2). In comparison, the mature stands' large trees comprised only 4% of all inventoried trees both in pine and spruce stands. The few large trees observed in mature stands (mostly in pine stands) (Fig. 2) may be explained by the history of forest management and stand structure[30,32,35]. Similar estimates have been obtained in Mediterranean forests in Spain (large trees (DBH > 50 cm) comprised 33% and 9% of all inventoried trees in old-growth and managed stands) regarding the large tree share of all trees, although the rotation age for managed stands was approximately twice as high compared to our studied stands[30]. For most of the old-growth spruce stands in this study, the estimated stand age was well above defined species-specific age limit in the European boreal zone (150 years), but for old-growth pine stands the stand age is approaching it (200 years), respectively[56]. Nevertheless, cored trees in our study also exhibited much older trees in the studied stands – up to 255 years for Scots pine and 249 years for Norway spruce in old-growth pine stands and up to 271 years for Norway spruce and 262 years for Scots pine in old-growth spruce stands.

Regarding the deadwood, there was a significant difference (58% and 85% for spruce and pine) in deadwood carbon stock between the old-growth and mature stages, but it was similar between dominant tree species. Similar estimates have been obtained also between old-growth and mature deciduous forest stands in the hemiboreal region[57]. At both age groups, in spruce stands, lying deadwood represented more than half (52% and 61% in mature and old-growth stage) of the total deadwood carbon stock. While in pine stands, dead standing trees stored slightly below 50% of the deadwood carbon and 38% of the total deadwood carbon constituted lying deadwood. Obtained differences regarding deadwood types reflected the differences in time necessary for spruce and pine to fall over (the majority of pines die standing and require times before falling over)[57], mainly due to slow decay rates and root system structure as reported for other tree species in previous studies[26]. The greatest share of the deadwood was moderately decomposed trees (Stages 2 and 3), which was also determined by wood properties of specific tree species, the time passed since the tree's death and tree size[14,57]. Moreover, previous studies in the hemiboreal region show that Scots pine can reach a notably older age compared to Norway spruce. The results from our study show that the carbon stock in tree biomass and deadwood continue to increase from the mature to the old-growth stage. Nevertheless, a high variability of tree biomass and deadwood carbon pools reflects differences in past natural disturbance regime and ongoing succession together with the dominant tree species effect—regeneration strategy and lifespan, as well as stand species composition, soil properties and application of silvicultural treatments in the managed stands[8,9,30,54].

Tree biomass carbon stock modelling

Old-growth stands in this study included forest stands where old trees are still the dominant forest element, potentially representing the upper limit of carbon pools characterising undisturbed hemiboreal old-growth

coniferous stands. The mean tree biomass carbon stocks differed significantly between pine- and spruce-dominated old-growth stands, with higher values observed for pine. The dominant tree species did have a visible effect on carbon storage in tree biomass in the old-growth stage; such a trend was also observed in mature stands, yet it to a smaller extent. This difference can largely be attributed to species-specific wood density, which is a key determinant of carbon content and carbon storage potential. Scots pine typically has higher carbon content compared to Norway spruce[45], resulting in greater carbon storage per unit of stem volume. In addition, structural differences, such as slower decomposition rates and longer lifespan of pine, may further contribute to higher maximum carbon stocks in pine-dominated stands[9]. Our first hypothesis was proven to be true - the large trees over 40 cm DBH in old-growth stands hold disproportionately large amounts of carbon (the mean 49–58% of tree biomass carbon stock) (Fig. 2). In mature stands, large trees account for significantly lower tree biomass carbon stock – 11% and 14% in pine and spruce stands, respectively. Rather similar values have been obtained from temperate forests in the western United States (42% of aboveground carbon in 3% of trees over 53 cm[58] and in tropical forests (~ 36% of above-ground biomass in ~ 3% trees over 70 cm[59]. Across all primary forest analyses in Europe showed, that large trees (DBH > 60 cm, representing 15% of the number of trees) accounted for 50% of total living biomass carbon stock[17]. Our obtained values are indeed higher than what has been reported so far, also in European old-growth temperate mountain forests (36% of carbon in 5% of trees over 100 cm[60], however, the applied DBH defining large trees varies between studies, as well as tree aboveground biomass carbon (biomass) instead of total tree carbon stock has been compared in reference studies. Apart for carbon storage, large old trees in such stands increase structural complexity and provide biodiversity hot spots, playing a crucial role in biodiversity maintenance[6,8,18].

Single-tree mortality impact on tree biomass carbon pool - sample plot analysis

Natural disturbances can fundamentally alter the size of tree biomass and deadwood carbon pools in old-growth stands[8,31]. Tree mortality is a complicated process affected by a variety of factors, and in unmanaged forests, many disturbances affect tree mortality simultaneously; therefore, it is difficult to predict and is highly uncertain[26,61]. There are multiple mechanisms/modes that can lead to the death of individual tree, including hydraulic limitation, wind-related damages, biotic attacks, tree size, and fire and this variation is challenging to assessing[24]. For example, recent observations of tree growth reduction are most likely linked to drought impact in forest ecosystem, which have shown high sensitivity in carbon uptake[11]. Although competition is often reduced for very large trees (up to 126 cm in DBH), previous studies have shown that survival rates may also decline in the largest size classes due to disturbance frequency and severity, which are expected to intensify under ongoing climate change[24]. Previous assessments of tree mortality in Europe analyzing uneven-aged Scots pine stands showed that as the tree DBH increases, the probability of a tree dying due to wind, insect or fungal damage also increases and the main cause of tree death is tree competition[53,61]. Internationally, there is a wide range of mortality models developed and available based on long-term permanent sample plots; however, we do not have a long enough time series with data from old-growth stands available to develop such a model so far. Nevertheless, it is highly likely that single-tree mortality[26] and species-specific effects[8,53,61] played a significant role in observed carbon storage differences and high variation between age groups according to the number of excluded trees. Therefore, to investigate whether and how the largest tree exclusion from sample plots affects the tree biomass carbon stock, we calculated the tree biomass carbon stock changes by excluding from 1 to 15 largest trees per sample plot. Overall, the mean initial tree biomass carbon stock per sample plot in old-growth stands was higher compared to two times younger mature stands. Exclusion of only the few largest trees caused significant mean tree biomass carbon stock reduction per sample plot in old-growth stands (Fig. 5), and it was more pronounced in spruce stands. In pine stands, significant differences in tree biomass carbon stock between old-growth and mature stands were obtained by excluding one to three of the largest trees per sample plot. However, when increasing the number of excluded trees, the difference in tree biomass carbon stock became similar between both the old-growth and mature stages, because initial tree distribution within diameter classes is different (Fig. 3a). Therefore, after excluding three of the largest trees, the rest of the trees in both age groups are of similar size. On the contrary, in spruce stands, significant differences of tree biomass carbon stock between old-growth and mature stands were obtained by excluding one to fifteen of the largest trees per sample plot, and the difference became even larger (Fig. 5c). The percentage reduction of tree carbon stock per sample plot was higher both in pine and spruce in old-growth stands. Significant ($p < 0.05$) differences between mature and old-growth stands were observed, excluding two to fifteen and one to fifteen trees per plot in pine and spruce stands, respectively (Fig. 5c). In old-growth stands, the exclusion of five to six of the largest trees in sample plots reduced carbon storage by just above 50%. Contrary, eight (for pine) and nine (for spruce) tree exclusions in sample plots reduced carbon storage slightly above 50%. Further assessment of tree size variability within the sample plots and the impact of tree exclusion showed that, in pine stands, the initial mean DBH was similar between mature and old-growth stages, whereas in spruce stands, a significant difference was already apparent before the exclusion of the largest trees within the sample plot. Our simulation demonstrated that, after excluding the eight largest trees from the sample plot, the mean DBH of the remaining trees differed significantly, and the decrease in mean DBH was overall sharper in old-growth compared to mature pine stands. In contrast, spruce stands already showed a significant difference in initial mean DBH between mature and old-growth stages, and this difference became more pronounced with the exclusion of additional trees. These results emphasize that tree size variability and the contribution of large individuals to stand structure differ between developmental stages and tree species. Also, observed differences were due to species differences with respect to the probability of mortality. It corresponds with previous studies well – Siipilehto et al.[62] were able to estimate that the probability of mortality is lowest in pine-dominated stands and increased the tolerance to increasing age (senescence). To assess the spatial distribution of dead trees, we calculated the mean UAI. It indicated an aggregated distribution of deadwood, with more than 50% of all deadwood having UAI > 0.5. The aggregated pattern appeared mainly

as a consequence of the live tree distribution pattern within the sample plots (also UAI > 0.5). The proportion of deadwood that formed regular patches in old-growth spruce stands was slightly higher than that of pine old-growth stands, suggesting that admixture of other tree species and spruce as a shade-tolerant species in spruce-dominated old-growth stands and also spruce composition in pine-dominated old-growth stands, are subjected to become more aggregated over time and with forest ageing[26,53]. Moreover, clustered pattern of smaller sized trees in old-growth forests has been often discovered, as younger trees filled up small gaps created by the death of single old trees[53]. The aggregated pattern of deadwood in old-growth coniferous stands found in our study is partly in accordance with the results obtained by Larson et al.[63], where dead trees showed a more aggregated distribution in old-growth stands compared to younger stands. Nevertheless, mortality in old-growth stands is density independent, but it is caused by a diverse set of agents and in many cases is indistinguishable from randomness, thus creating and maintaining spatial heterogeneity in the old-growth stage[63]. Therefore, large carbon storage in tree biomass in old-growth stands is most likely possible only while single-tree mortality occurs. Furthermore, this demonstrating that different natural disturbance regimes generate fluctuating trade-offs in provided forest functions (structure-based biodiversity potential, carbon storage) in long-term perspective[22]. Further monitoring of natural disturbance patterns and their effects on tree (live and dead) carbon pools is essential to elaborate the conclusions on potential long-term carbon storage in old-growth forests, also allowing for better management of deadwood resources[23].

Conclusion

The present study shows that stand density decreases with increasing tree size (larger DBH and stand age), and overall, tree density was much lower in old-growth stands. Old-growth stands in this study included forest stands where old trees are still the dominant forest element, potentially representing the upper limit of carbon pools characterising undisturbed hemiboreal old-growth coniferous stands. However, the mature stand ages do not correspond to a specific situation; for example, when a particular tree species reaches its maximum growth rate. Nevertheless, concerning single-tree mortality per sample plot, results revealed that the exclusion of a few largest trees caused a significant decrease of the mean tree biomass carbon stock between old-growth and mature stands, and the percentage reduction was higher both in old-growth pine and spruce in stands. However, carbon loss was modeled based on the exclusion of the largest trees not considering tree density and/or other impacting factors. The current study represents a starting point for further research and analysis of stand structural differences and their impact on long-term carbon storage dynamics in old-growth stands in the hemiboreal forest zone.

Data availability

The datasets generated during and/or analyzed during the current study are available from the corresponding author.

Received: 5 March 2025; Accepted: 21 January 2026
Published online: 28 January 2026

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Acknowledgements
The authors would like to thank to Annele Baltmane for her assistance with English language editing and grammar checking of the manuscript.

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Conceptualization: L.Ķ., Ā.J.; Methodology: Ā.J., L.Ķ., D.E., I.J.; Data collection: I.J., L.Ķ.; Formal analysis and investigation: L.Ķ., D.E., Ā.J.; Writing - original draft preparation: L.Ķ.; Writing - review and editing: L.Ķ., Ā.J., D.E.; Funding acquisition: Ā.J.; Supervision: Ā.J.

Funding
This study was supported by the European Regional Development Fund project "Development of a decision support tool integrating information from old-growth semi-natural forest for more comprehensive estimates of carbon balance" (No. 1.1.1.1/19/A/130) and the Latvia Council of Science national research programme project: "Forest4LV – Innovation in Forest Management and Value Chain for Latvia's Growth: New Forest Services, Products and Technologies" (No. VPP-ZM-VRIIILA-2024/2–0002).

Declarations

Competing interests
The authors declare no competing interests.

Consent for publication
All authors gave their informed consent to this publication and its content.

Additional information
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